LangGraph-M-H-Set2

20 questions | Medium to hard | Choose one answer per question

01. Several nodes need a request-scoped database client. It is a live connection object, not business data to checkpoint. Where should the graph receive this dependency?

A. In a normal state field so every checkpoint serializes the connection.

B. Through runtime context, supplied for the invocation and accessed by nodes.

C. In the messages channel as a system message containing the connection.

D. As a new graph node name derived from the connection identifier.

02. A graph needs internal evidence and review_notes, but ordinary invoke() results should expose only final_report. Which design provides this output contract?

A. Attach a checkpointer, which automatically hides every intermediate field.

B. Remove evidence and review_notes from all state schemas before execution.

C. Rename the internal fields with an underscore and rely on automatic filtering.

D. Keep the internal state schema and define an output schema containing final_report.

03. The parent state contains question and answer. A reusable child graph accepts query and returns response, with no shared keys. Which integration preserves both interfaces?

A. Wrap the child invocation in a node that maps parent input and child output.

B. Give the parent and child the same thread_id to translate their field names.

C. Add a reducer that renames all parent fields before every graph step.

D. Add the child directly and assume question is automatically renamed to query.

04. A node inside a child graph must hand control to finalize in its nearest parent graph. finalize is not a child node. Which routing instruction identifies the correct graph?

```
A. Command(goto="finalize")
B. Command(update={"graph": "parent", "next": "finalize"})
C. Command(goto="finalize", graph=Command.PARENT)
D. Send("finalize", {"graph": "parent"})
```

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05. A cached retrieval node receives tenant_id and query, but its custom cache key uses only query. Two tenants submit identical text against different corpora. What is the best correction?

A. Include tenant and relevant corpus/version identity in the cache key.

B. Keep the key unchanged and distinguish tenants only with trace tags.

C. Use a longer cache TTL so both requests are served consistently.

D. Keep the key unchanged and enable checkpointing on the graph.

06. Nodes A and B run in the same superstep with count initially 0. A returns count=1. B reads its supplied state after A has finished. With no shared-object mutation, what does B read?

A. 1, because the fastest completed node publishes immediately.

B. 0, because node updates become visible in the next step.

C. Either 0 or 1, depending only on operating-system scheduling.

D. An error, because parallel nodes cannot read the same field.

07. A custom numeric merge is intended to give the same answer when partial results are grouped differently. Which analysis of the candidate below is correct?

```
merge = lambda a, b: a - b
merge(merge(10, 3), 2)
merge(10, merge(3, 2))
```

A. It is associative because subtraction accepts two numeric arguments.

B. It is associative as long as the graph has a checkpointer.

C. It becomes associative if all workers use the same Python type.

D. It is not associative: the two displayed groupings produce 5 and 9.

08. Parallel workers produce numbered report sections. The final report must follow the requested section order, regardless of worker timing or graph topology. Which design is most robust?

A. Include a section index in each result and sort during assembly.

B. Concatenate results in the order workers happen to finish.

C. Rename workers alphabetically and treat that as the report specification.

D. Assign the same section index to every result and concatenate them.

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09. A checkpoint has audit=["old"], and audit uses operator.add for list concatenation. An operator calls graph.update_state(config, {"audit": ["corrected"]}). What does the new checkpoint contain?

A. audit=["corrected"], because update_state bypasses reducers.

B. audit=[], because manual updates clear accumulated lists first.

C. audit=["old", "corrected"], because the field reducer still applies.

D. audit=["old"], because checkpointed values cannot be edited.

10. In the linear graph shown, an operator edits a saved draft using update_state(..., as_node="draft"). After resuming from the returned config, what does this attribution mean?

```
START -> retrieve -> draft -> evaluate -> END
```

A. Rename the edited state field to draft and keep routing unchanged.

B. Run the draft function immediately inside update_state, then stop.

C. Jump directly to END because a human supplied the update.

D. Treat the update as output from draft and continue toward evaluate.

11. You replay from a checkpoint immediately before an LLM drafting node. There is no node cache. How should you interpret the replayed draft?

A. It must be identical because a checkpoint fixes every future model response.

B. It may differ because nodes after the selected checkpoint execute again.

C. It is identical unless you change the thread_id before replaying.

D. It is generated without another model call by reading the old final result.

12. Parallel nodes A and B share a superstep. A finishes and its pending writes are durably saved; B fails. You resume that same failed execution with a supported checkpointer. What recovery behavior is expected?

A. Rerun both nodes because a partial superstep can never preserve useful work.

B. Skip B and treat the whole superstep as successfully completed.

C. Reuse A's saved writes and retry the unfinished or failed work in B.

D. Discard A's output and substitute the previous checkpoint's value permanently.

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13. A retrieval node sometimes raises ConnectionError during a temporary outage. It also raises ValueError for malformed document IDs. Which policy best avoids pointless repeated attempts?

A. Use bounded retries for ConnectionError and surface ValueError for correction.

B. Catch both exceptions and return an empty successful result.

C. Retry both exception types forever because retrieval is read-only.

D. Apply the same delay to all errors and disable attempt limits.

14. Saved executions are paused before the node named legal_review. A new deployment deletes that node and introduces review_v2. What is the main compatibility issue?

A. The checkpointer will always infer the rename from similar node implementations.

B. Saved execution positions may still reference legal_review; preserve compatibility or migrate those runs.

C. Changing the name affects visualization only, so paused executions are unaffected.

D. Deleting the node automatically marks every pending review as approved.

15. A developer wraps interrupt() in a bare except block that returns {"status": "failed"}. The caller never receives the intended approval pause. What is the best explanation?

A. The interrupt payload must always be a dictionary rather than a string.

B. A bare except block creates an additional checkpoint automatically.

C. Calling interrupt inside any Python function is unsupported.

D. The handler can swallow the control-flow exception that signals the pause.

16. A node originally asks for name and then age using two interrupt calls. After the first pause, a deployment reverses their order before resumption. Why is this unsafe?

A. Resume values are matched by English question text, which may be ambiguous.

B. Only the last interrupt in any node is ever persisted.

C. Resume values are matched by call order within the task, so the changed order can misapply them.

D. Every interrupt always starts a new independent thread automatically.

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17. An AIMessage requests two calls to the same search tool with IDs call_a and call_b. Your custom tool-execution node returns ToolMessages. What must distinguish the two results?

A. Different human-readable tool names, even though both calls use search.

B. The same tool_call_id, because the tool function name is the same.

C. Each result's tool_call_id must match its corresponding requested call ID.

D. Their order in the Python list is sufficient; identifiers are unnecessary.

18. A trimming step removes an AIMessage containing a tool call but keeps its ToolMessage result. The next model request is rejected for invalid message history. What should the trimming policy do?

A. Change the orphaned result's ID to a random value and retry.

B. Convert the orphaned ToolMessage into a system message without inspecting it.

C. Keep only ToolMessages because their factual content is sufficient.

D. Preserve or remove the related tool-call/result exchange coherently.

19. A long-running node should emit "30% complete" to the UI without storing that progress message as business state. Using the stream-mode API, which pattern fits?

A. Print the message to stdout and assume graph.stream forwards every print.

B. Emit through get_stream_writer() and consume the custom stream mode.

C. Put progress in the final output schema and wait for the whole run to finish.

D. Return a state update after each internal loop iteration without ending the node.

20. A graph uses max_concurrency=4 for scheduled work, yet its model provider reports token-per-minute limit errors. Which conclusion is most accurate?

A. Concurrency limits simultaneous work; token-rate control needs separate budgeting or throttling.

B. Four concurrent tasks always imply no more than four tokens per minute.

C. The error proves that max_concurrency cannot affect execution at all.

D. Changing thread_id on every call automatically resets the provider's token quota.